

Solve the following problems:

- (1) Consider a computer system with Poisson job-arrival stream at an average of 3 per minute. Determine the probability that in any one-minute interval there will be:
 - a. 0 jobs
 - b. Exactly 3 jobs
 - c. At most 4 arrivals.
- (2) In a random sample of 1000 persons from town A, 480 are found to be consumers of wheat. In a sample of 800 from town B, 420 are found to be consumers of wheat. At the 1% level of significance, perform a hypothesis test to figure out a significant difference between town A and town B, in the context of the proportion of wheat consumers are concerned.
 - a. Use the critical value approach.
 - b. Perform the test using the p -value approach.